

BUTLER AI — COMPANION SERVER & REMOTE RELAY GUIDE

Version 20.1.0 (OSS Release) · Secure PC Automation & Zero-Knowledge Tunneling

1. Python Companion Server Architecture

The Butler AI Python companion server (`butler_server_v20_1_0_OSS.py`) runs locally on your Windows, macOS, or Linux workstation. It provides secure REST endpoints for PC telemetry, memory storage, script workshop management, and local Ollama model orchestration. Built on FastAPI and Uvicorn, the server adheres to a strict **fail-closed privacy circuit**: no telemetry, diagnostics, or chat logs are ever transmitted to external servers.

2. Hardened Vault & E2E Encryption

Local memory and paired session tokens are protected by a **Hardened Memory Vault** implementing AES-256-GCM encryption paired with PBKDF2-HMAC-SHA256 key derivation. Brute-force attempts against the 6+ digit PIN trigger an automatic fail-closed lockout after 5 consecutive failures, isolating sensitive data from unauthorized physical or remote access.

3. Secure Remote Relay Pass (\$5 Add-On)

When operating outside your home Wi-Fi over cellular networks or foreign hotspots, the **Secure Remote Relay Pass** enables encrypted out-of-home control without compromising zero-knowledge guarantees:

- **Curve25519 ECDH Key Exchange:** Establishes a secure cryptographic session between mobile app and PC companion.
- **AES-256-GCM Envelope Encryption:** All automation commands and script payloads are fully encrypted before touching intermediary relay nodes.
- **Blind Forwarder Relay:** Relay nodes forward opaque binary blobs with zero decryption capability, preventing third-party interception.
- **Lifetime Value:** A one-time \$5.00 USD purchase unlocks lifetime encrypted remote relay access across personal devices.

4. Quick Start & Operation

Step	Command / Action	Description
1	Install Dependencies	Run <code><code>pip3 install fastapi uvicorn pydantic</code></code>
2	Start Server	Run <code><code>python3 butler_server_v20_1_0_OSS.py</code></code>
3	Pair Mobile App	Scan the terminal QR code or enter IP manually
4	Automate & Secure	Execute scripts, monitor CPU/RAM, and enjoy bulletproof local control